

Confidence Intervals: Estimating the Population Mean

Statistics Worksheet · Grade 11–12

Name: _____

Date: _____

Learning Objectives

- Understand what a confidence interval is and how it estimates the true population mean
- Calculate confidence intervals using sample statistics and z-scores
- Interpret confidence levels and explain what they mean in context

Problems

1. A researcher wants to estimate the average weight of rabbits in a large population. She knows the population mean is unknowable. Which of the following best describes a confidence interval?

2. A sample of 30 rabbits has a mean weight of 3.2 lb and a population standard deviation of 0.8 lb. Identify the point estimate of the population mean.

$$\bar{x} = 3.2 \text{ lb}, \quad \sigma = 0.8 \text{ lb}, \quad n = 30$$

3. Three samples were taken from a rabbit population. Sample 1 ($n = 30$) gave $\bar{x} = 3.2$ lb, Sample 2 ($n = 245$) gave $\bar{x} = 4.2$ lb, and Sample 3 ($n = 326$) gave $\bar{x} = 5.9$ lb. Which sample is most likely to produce the most reliable estimate of the population mean, and why?

4. Calculate the standard error of the mean for a sample of 50 rabbits if the population standard deviation is 1.4 lb.

$$SE = \frac{\sigma}{\sqrt{n}}, \quad \sigma = 1.4, \quad n = 50$$

5. A confidence interval for rabbit weights is reported as (3.5, 5.5) lb at an 80% confidence level. What is the margin of error for this interval?

$$CI = (3.5, 5.5)$$



6. The table below shows results from repeated samples of size 50 from a rabbit population. The confidence interval was found to be (3.5, 5.5) lb. Determine which sample means fall **INSIDE** the confidence interval and which fall **OUTSIDE**.

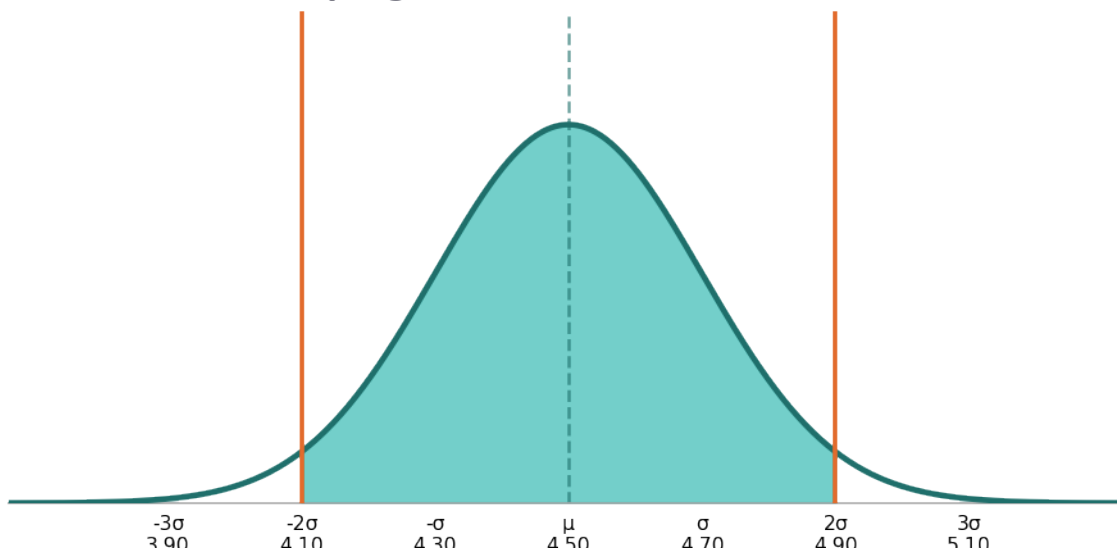
Sample	Sample Mean (lb)	Inside CI (3.5, 5.5)?
1	4.1	
2	5.2	
3	3.1	
4	4.8	
5	5.7	

7. A sample of 64 rabbits has a mean weight of 4.5 lb. The population standard deviation is known to be 1.6 lb. Construct a 95% confidence interval for the true population mean. Use z-star = 1.96.

$$CI = \bar{X} \pm z^* \cdot \frac{\sigma}{\sqrt{n}}$$

8. The shaded region below represents the area under a normal distribution for a sample mean. The population of rabbit weights has a mean of 4.5 lb and a standard deviation of 1.6 lb with $n = 64$. Find the probability that a randomly selected sample mean falls between 4.1 and 4.9 lb.

Sampling Distribution of $\bar{x}$ ($n = 64$)

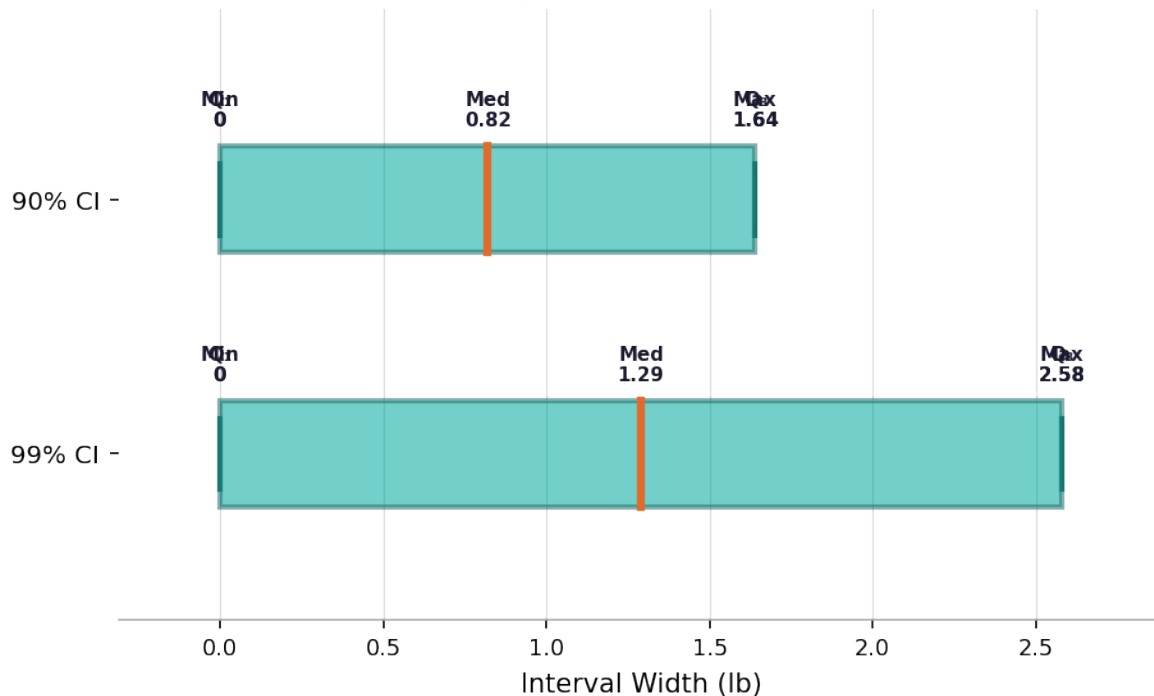


9. A researcher wants to estimate the mean weight of rabbits within a margin of error of 0.25 lb at a 99% confidence level. The population standard deviation is 1.5 lb. Use z-star = 2.576. What minimum sample size is required?

$$n = \left(\frac{z^* \cdot \sigma}{E} \right)^2$$

10. Two researchers study the same rabbit population. Researcher A uses a 90% confidence level and Researcher B uses a 99% confidence level, both with the same sample size and standard deviation. The box plot below shows the widths of their confidence intervals. Researcher A reports CI width = 1.64 lb and Researcher B reports CI width = 2.58 lb. Explain which interval is wider, why this occurs, and what trade-off exists between confidence level and interval precision.

CI Width Comparison by Confidence Level



Confidence Intervals: Estimating the Population Mean — Answer Key

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Answer Key

1. Answer: A range of values computed from a sample that is likely to contain the true population mean

- A confidence interval is NOT a single value — it is a range (lower bound, upper bound).
- It is calculated from sample data and is designed to capture the true population mean with a stated level of confidence.
- Answer: A range of values computed from a sample that is likely to contain the true population mean.

2. Answer: Point estimate = 3.2 lb

- The point estimate of the population mean (μ) is simply the sample mean ($\bar{x}$).
- $\bar{x} = 3.2$ lb
- Therefore, the point estimate is 3.2 lb.

3. Answer: Sample 3 ($n = 326$); larger sample size reduces sampling error and produces a more reliable estimate

- As sample size increases, the standard error of the mean decreases: $SE = \sigma / \sqrt{n}$.
- Larger $n \rightarrow$ smaller $SE \rightarrow$ the sample mean is closer to the true population mean.
- Sample 3 has the largest n (326), so it provides the most reliable estimate.

4. Answer: $SE \approx 0.198$ lb

- $SE = \sigma / \sqrt{n} = 1.4 / \sqrt{50}$
- $\sqrt{50} \approx 7.071$
- $SE = 1.4 / 7.071 \approx 0.198$ lb

5. Answer: Margin of error = 1.0 lb

- The margin of error (E) is half the width of the confidence interval.
- Width = Upper bound – Lower bound = $5.5 - 3.5 = 2.0$
- $E = \text{Width} / 2 = 2.0 / 2 = 1.0$ lb

6. Answer: Inside: Samples 1, 2, 4; Outside: Samples 3, 5

Sample	Sample Mean (lb)	Inside CI (3.5, 5.5)?
1	4.1	Yes
2	5.2	Yes
3	3.1	No



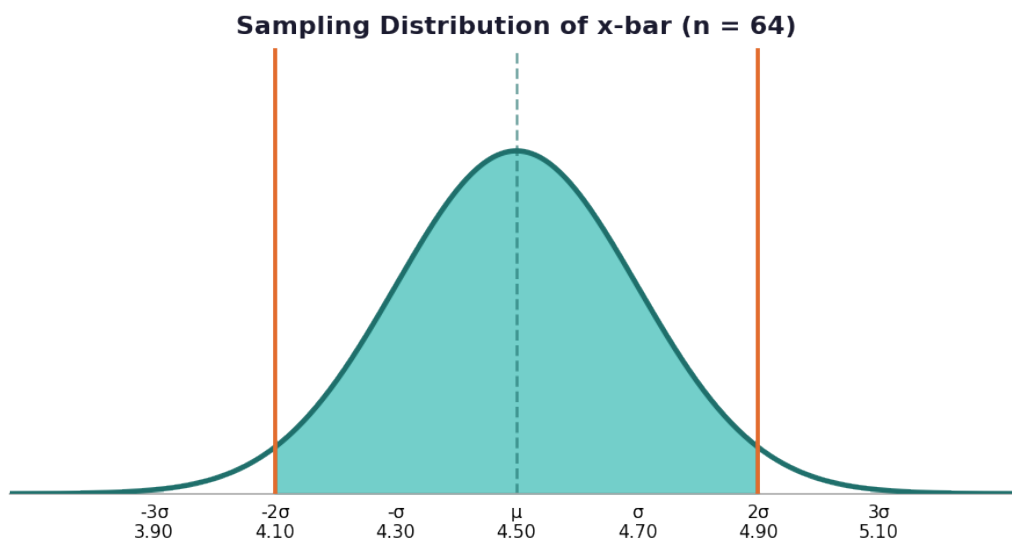
Sample	Sample Mean (lb)	Inside CI (3.5, 5.5)?
4	4.8	Yes
5	5.7	No

- Check each mean: is $3.5 \leq \bar{x} \leq 5.5$?
- 4.1: Yes (inside) | 5.2: Yes (inside) | 3.1: No (below 3.5) | 4.8: Yes (inside) | 5.7: No (above 5.5)
- 3 out of 5 samples fall inside the interval.

7. Answer: 95% CI: (4.108, 4.892) lb

- $SE = \sigma / \sqrt{n} = 1.6 / \sqrt{64} = 1.6 / 8 = 0.2$
- Margin of error $E = z^* \times SE = 1.96 \times 0.2 = 0.392$
- Lower bound = $4.5 - 0.392 = 4.108$
- Upper bound = $4.5 + 0.392 = 4.892$
- 95% CI: (4.108, 4.892) lb

8. Answer: $P \approx 0.9545$ (95.45%)



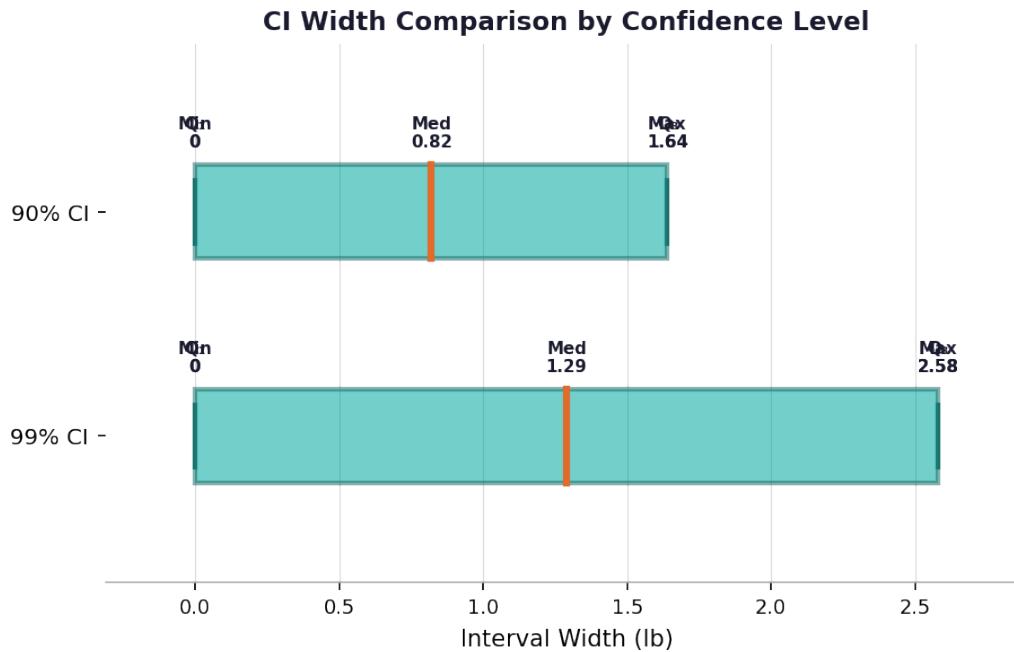
- $SE = \sigma / \sqrt{n} = 1.6 / 8 = 0.2$
- $z_1 = (4.1 - 4.5) / 0.2 = -2.00$
- $z_2 = (4.9 - 4.5) / 0.2 = 2.00$
- $P(-2 < z < 2) \approx 0.9545$
- There is approximately a 95.45% chance the sample mean falls between 4.1 and 4.9 lb.

9. Answer: Minimum n = 239

- Use the sample size formula: $n = (z^* \times \sigma / E)^2$
- $n = (2.576 \times 1.5 / 0.25)^2$
- $= (3.864 / 0.25)^2$
- $= (15.456)^2$
- $= 238.887$
- Always round UP to the next whole number: $n = 239$



10. Answer: Researcher B's 99% CI is wider. Higher confidence requires a larger z^* , which increases the margin of error, producing a wider but less precise interval.



- Margin of error $E = z^* \times (\sigma / \sqrt{n})$. For 90%, $z^* = 1.645$; for 99%, $z^* = 2.576$.
- Since z^* is larger for 99%, the margin of error is larger, making the CI wider.
- Researcher B's interval (2.58 lb wide) is wider than Researcher A's (1.64 lb wide).
- Trade-off: A higher confidence level gives more certainty that the interval captures μ , but the interval is less precise (wider range).
- A lower confidence level gives a narrower, more precise interval, but with less certainty of capturing μ .

